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Unit “ 3 : Session “ 6 : SLO “ 1 | Stack Applications “” Recursion

■ I. Fill in the Blanks

- In recursion, each function call is stored in the STACK (call stack).
- The LIFO principle is followed by stacks.
- Stack is used internally to implement RECURSIVE function calls.
- Every recursive function must have a BASE condition to avoid infinite calls.
- The process of returning from a recursive function happens in the REVERSE order of calls.
- Each recursive call adds a new ACTIVATION RECORD (stack frame) to the stack.
- FACTORIAL (Tower of Hanoi / Fibonacci) is a classic example of a problem solved using recursion and stacks.
- The return address and local variables are stored in the function's ACTIVATION RECORD (stack frame).
- The system stack helps in managing NESTED (multiple) function calls during recursion.
- When the base case is reached, the stack starts to UNWIND (pop frames).